

Question ID: 038d87d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

A neighborhood consists of a **2**-hectare park and a **35**-hectare residential area. The total number of trees in the neighborhood is **3,934**. The equation **$2x + 35y = 3,934$** represents this situation. Which of the following is the best interpretation of x in this context?

Answer

- A. The average number of trees per hectare in the park
- B. The average number of trees per hectare in the residential area
- C. The total number of trees in the park
- D. The total number of trees in the residential area

Correct Answer: A

Rationale

Choice A is correct. It's given that a neighborhood consists of a **2**-hectare park and a **35**-hectare residential area and that the total number of trees in the neighborhood is **3,934**. It's also given that the equation **$2x + 35y = 3,934$** represents this situation. Since the total number of trees for a given area can be determined by taking the number of hectares times the average number of trees per hectare, this must mean that the terms **$2x$** and **$35y$** correspond to the number of trees in the park and in the residential area, respectively. Since **$2x$** corresponds to the number of trees in the park, and **2** is the size of the park, in hectares, **x** must represent the average number of trees per hectare in the park.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.